

task1:Retrieve all airline names in uppercase.

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with the 'airline' table selected under the 'public' schema. The main query editor contains the following SQL query:

```
1 select upper(airline_name) from airline
2
3 ;
```

The 'Data Output' tab is active, showing the results of the query. The results are displayed in a table with one column, 'upper text', and 13 rows of data. The status bar at the bottom indicates 'Total rows: 50' and 'Query complete 00:00:00.227'.

upper text
IPC
PDN
KLE
KHS
YLQ
NGL
0
QIG
NQX
SOZ
IYA
KOQ
IFH

task2:Replace any occurrence of the word "Air" in airline names with "Aero".

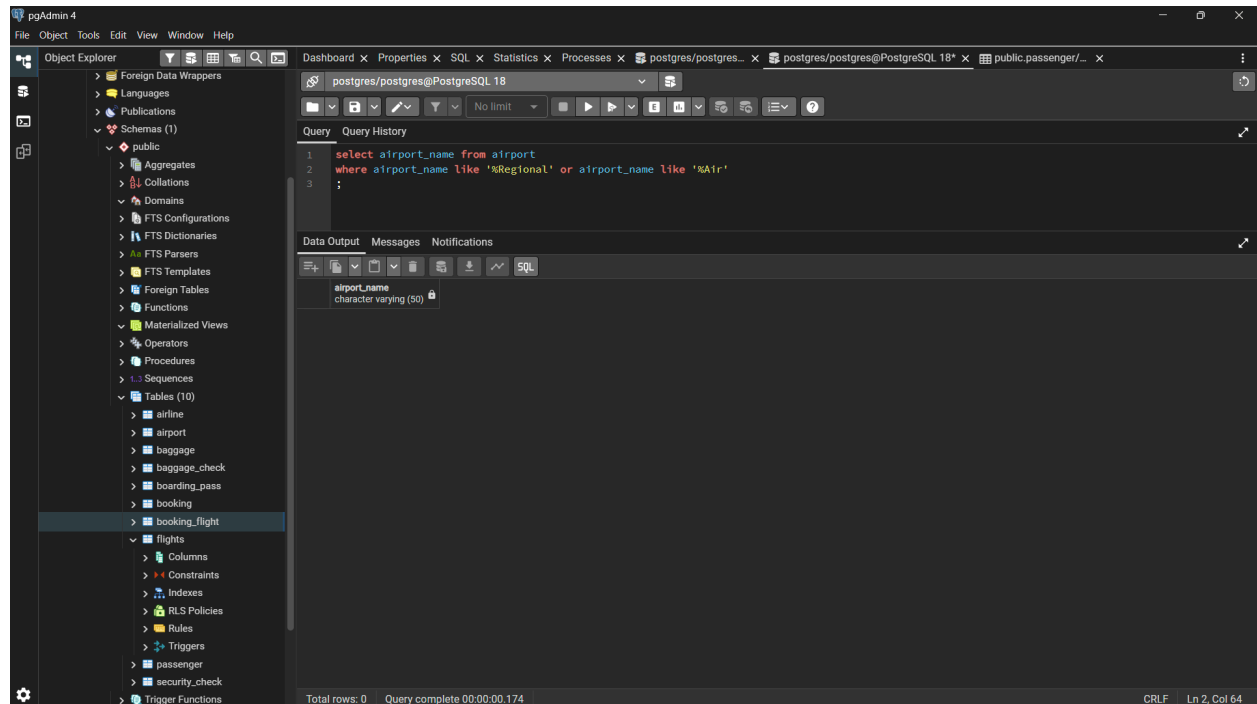
The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with the 'airline' table selected under the 'public' schema. The main query editor contains the following SQL query:

```
1 update airline
2 set airline_name = 'Aero'
3 where airline_name like '%B_';
```

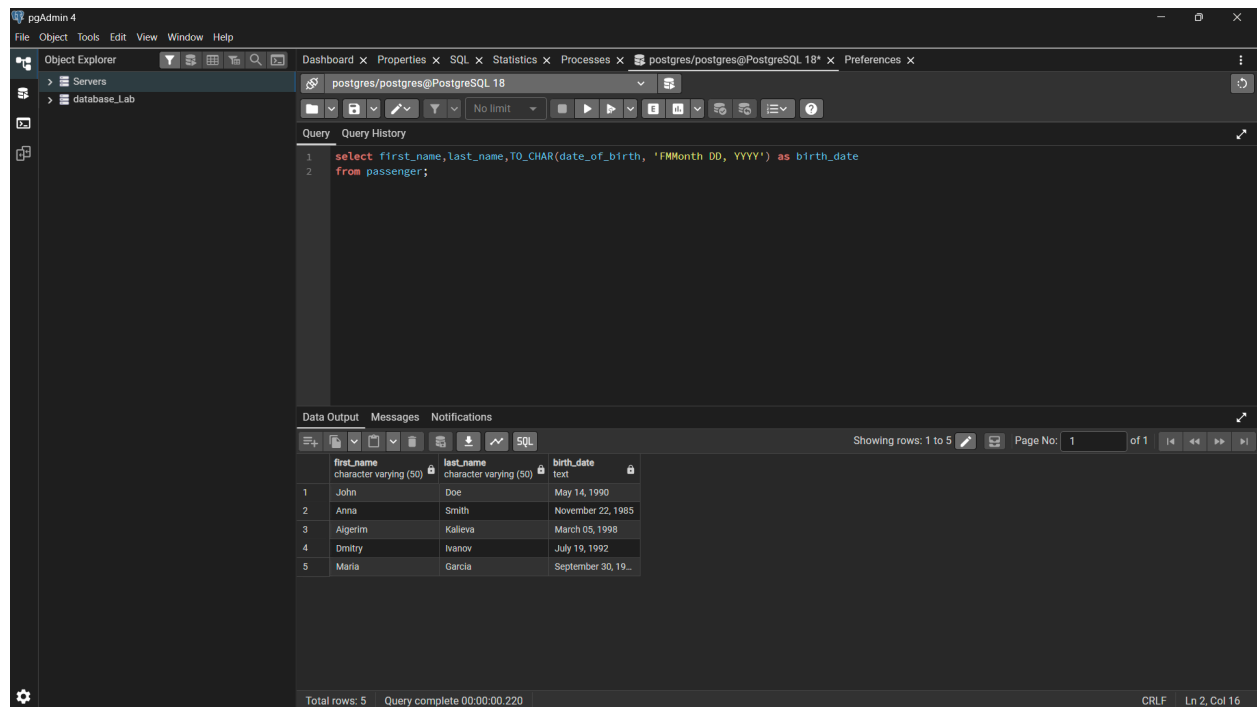
The 'Data Output' tab is active, showing the results of the query. The results are displayed in a table with columns: 'airline_id [PK] integer', 'airline_code character varying (50)', 'airline_name character varying (50)', 'airline_country character varying (50)', 'created_at date', and 'update_at date'. The status bar at the bottom indicates 'Total rows: 50' and 'Query complete 00:00:00.137'.

airline_id [PK] integer	airline_code character varying (50)	airline_name character varying (50)	airline_country character varying (50)	created_at date	update_at date
45	47	AYDN	DNJ	Russia	2024-01-22
46	48	ZMMG	MXW	China	2023-11-19
47	49	[null]	PQS	Poland	2023-04-21
48	33	KBVU	Aero	Russia	2023-03-31
49	45	HCMV	Aero	Sweden	2024-02-18
50	50	KBVH	Aero	Portugal	2023-09-08

Task 4 Retrieve airports that contain the word "Reginal" and "Air" in their names.



Task 5 Retrieve passenger names and format their birth dates as 'Month DD, YYYY'.



Task 6-7: Find flight numbers that have been delayed based on the actual arrival time.

pgAdmin 4

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Query Query History

```

1 select flight_id,flight_no , scheduled_departure, actual_departure from flights
2 where scheduled_departure < actual_departure;
3 --select * from flights;

```

Data Output Messages Notifications

Showing rows: 1 to 500 Page No: 1 of 1

	flight_id [PK] integer	flight_no character varying (50)	scheduled_departure date	actual_departure date
1	2	US-NM	2023-07-21	2024-02-09
2	3	FI-OL	2023-03-29	2024-02-21
3	5	RO-DJ	2023-07-03	2023-11-18
4	6	CA-SK	2023-07-07	2024-02-19
5	7	AU-TAS	2023-10-12	2023-12-04
6	9	IN-OR	2023-05-18	2023-06-17
7	11	TH-S7	2023-03-28	2024-02-06
8	12	CA-NL	2023-03-22	2023-04-28
9	14	US-IN	2023-07-24	2024-03-04
10	17	ES-AN	2023-09-10	2023-11-23
11	21	EE-S9	2023-10-12	2023-10-28

Total rows: 500 Query complete 00:00:00.348

Successfully run. Total query runtime: 00:00:00.348 CRLF Ln 1, Col 1

Task 8: Show airlines from any of: ('France','Portugal','Poland') created between '2023-11-01' and '2024-03-31'.

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Query Query History

```

1 Select airline_country,created_at,update_at from airline
2 where airline_country = 'France' or airline_country = 'Portugal' or airline_country = 'Poland'
3 and created_at>= '2023-11-01' and created_at < '2024-03-31'
4 ;

```

Data Output Messages Notifications

Showing rows: 1 to 9 Page No: 1 of 1

	airline_country character varying (50)	created_at date	update_at date
1	France	2023-11-18	2024-02-18
2	Portugal	2024-01-09	2023-11-09
3	Poland	2023-12-24	2023-10-30
4	Portugal	2024-03-11	2023-08-24
5	France	2023-09-04	2023-06-19
6	Poland	2023-11-03	2023-12-06
7	Poland	2023-11-24	2023-10-07
8	France	2023-11-21	2023-05-30
9	Portugal	2023-09-08	2023-12-25

Total rows: 9 Query complete 00:00:00.159

CRLF Ln 2, Col 86

Task 9 :Find top3 overweighted baggage with more than 25kg.

The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query is:

```
1 SELECT baggage_id, weight_in_kg
2 from baggage
3 where weight_in_kg > '25'
4 order by weight_in_kg desc
5 limit 3;
```

The Data Output tab shows the following results:

baggage_id	weight_in_kg
321	49.80
61	49.57
102	49.44

At the bottom, a green status bar indicates: "Successfully run. Total query runtime: 178 msec. 3 rows affected."

Task 10:Find the youngest passengers' full name.

The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query is:

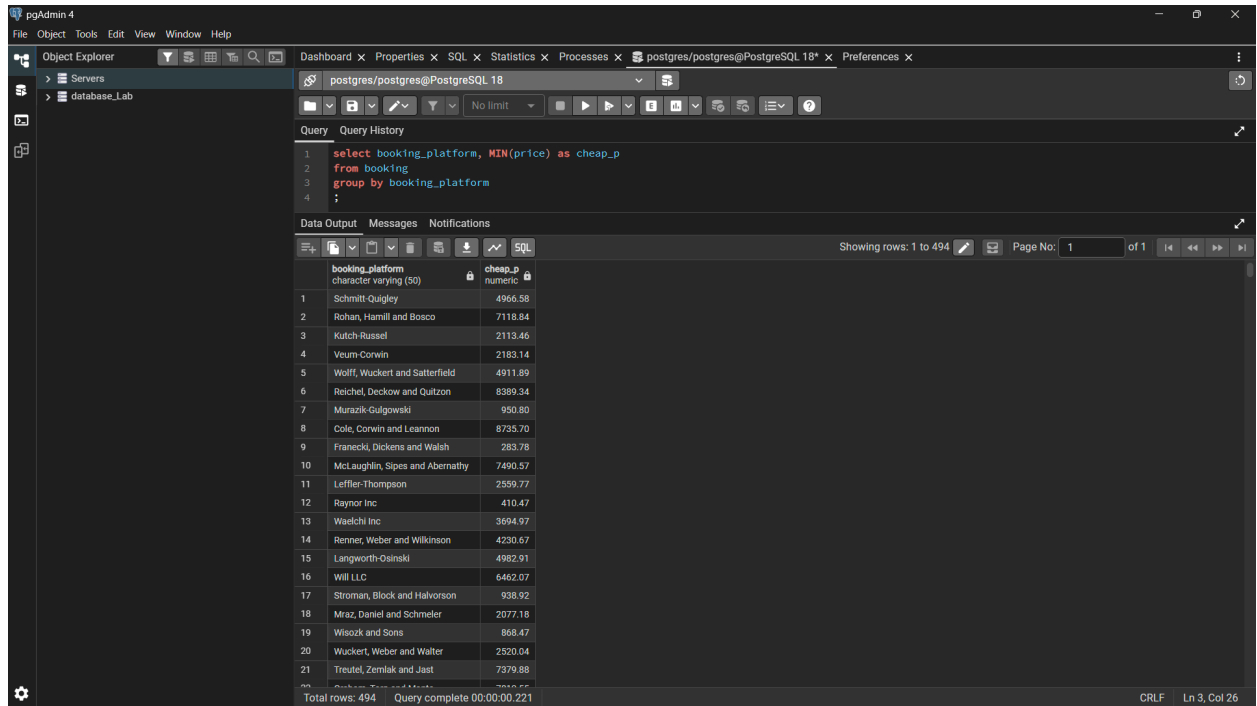
```
1 select * from passenger
2 order by date_of_birth desc
3 limit 1;
```

The Data Output tab shows the following results:

passenger_id	first_name	last_name	date_of_birth	gender	country	created_at	update_at
3	Algerin	Kalleva	1998-03-05	Female	Kazakhstan	2023-03-01	2023-08-11

At the bottom, a green status bar indicates: "Successfully run. Total query runtime: 161 msec. 1 rows affected."

Task 11: Find the cheapest booking price on each booking platform



The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query is:

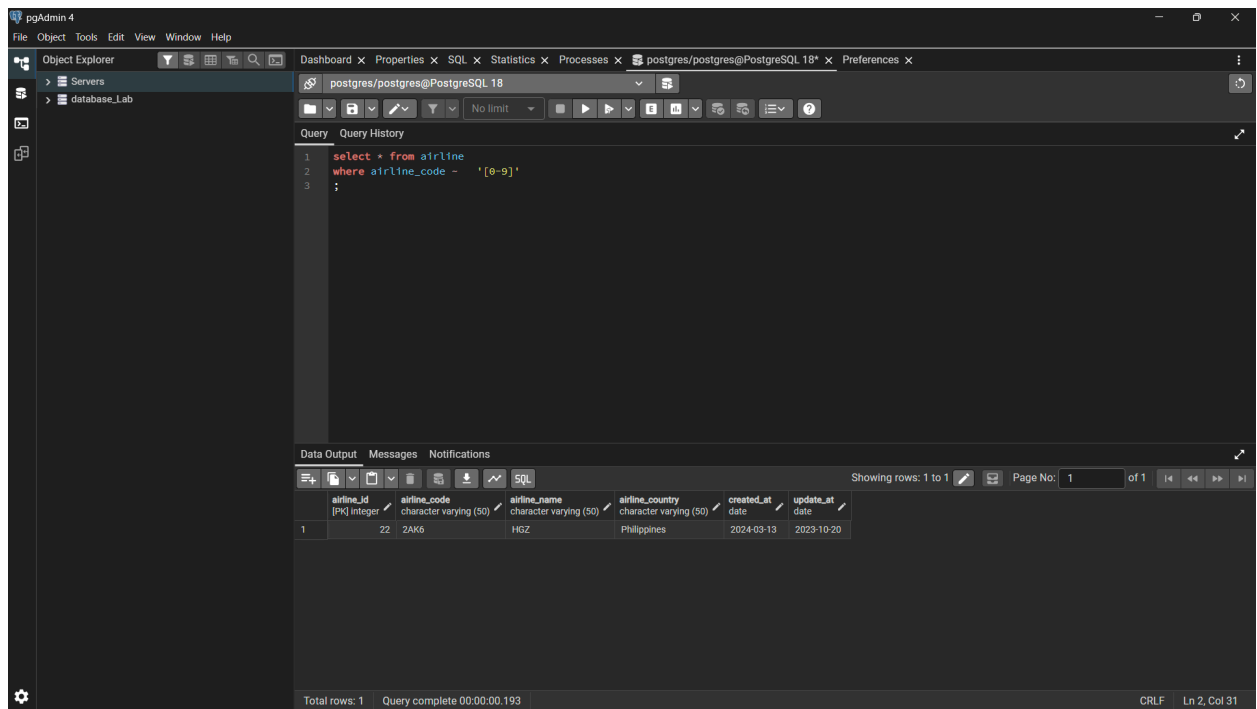
```
1 select booking_platform, MIN(price) as cheap_p
2 from booking
3 group by booking_platform
4 ;
```

The Data Output tab shows the results of the query, displaying the booking platform and the minimum price (cheap_p) for each platform. The results are as follows:

booking_platform	cheap_p
Schmitt-Quigley	4966.58
Rohan, Hamill and Bosco	7118.84
Kutich-Russel	2113.46
Vcum Corwin	2183.14
Wolff, Wuckert and Satterfield	4911.89
Reichel, Deckow and Quitzon	8389.34
Murazik-Gulgowski	950.80
Cole, Corwin and Leannon	8735.70
Franecki, Dickens and Walsh	283.78
McLaughlin, Sipes and Abernathy	7490.57
Leffler-Thompson	2559.77
Raynor Inc	410.47
Waelchi Inc	3694.97
Renner, Weber and Wilkinson	4230.67
Langworth-Osinski	4982.91
Will LLC	6462.07
Stroman, Block and Halvorson	938.92
Mraz, Daniel and Schmeler	2077.18
Wisozk and Sons	868.47
Wuckert, Weber and Walter	2520.04
Treutel, Zemlak and Jast	7379.88

Total rows: 494 Query complete 00:00:00.221

task12: Return airlines whose airline_code contains a digit.



The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query is:

```
1 select * from airline
2 where airline_code ~ '[0-9]'
3 ;
```

The Data Output tab shows the results of the query, displaying the airline details for the first row. The results are as follows:

airline_id	airline_code	airline_name	airline_country	created_at	update_at
22	2AK6	HGZ	Philippines	2024-03-13	2023-10-20

Total rows: 1 Query complete 00:00:00.193

Task 13: List the top5 most recently created airlines.

pgAdmin 4

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Query History

```

1 select * from airline
2 order by created_at desc
3 limit 5
4 ;

```

Data Output Messages Notifications

Showing rows: 1 to 5 Page No: 1 of 1

	airline_id [PK] integer	airline_code character varying (50)	airline_name character varying (50)	airline_country character varying (50)	created_at date	update_at date
1	9	KNOX	NQX	Venezuela	2024-03-16	2023-09-16
2	40	LTBD	CIL	Argentina	2024-03-15	2024-01-29
3	22	2AK6	HGZ	Philippines	2024-03-13	2023-10-20
4	24	ZSRG	RUG	Portugal	2024-03-11	2023-08-24
5	45	HCMV	Aero	Sweden	2024-02-18	2023-12-10

Total rows: 5 Query complete 00:00:00.209 CRLF Ln 3, Col 8

task 3: Find all flight numbers that coordinates with both airline 1 and airline 2.

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Query History

```

1 select flight_no
2 from flights
3 where airline_id =1
4
5 intersect
6
7 select flight_no
8 from flights
9 where airline_id =2;

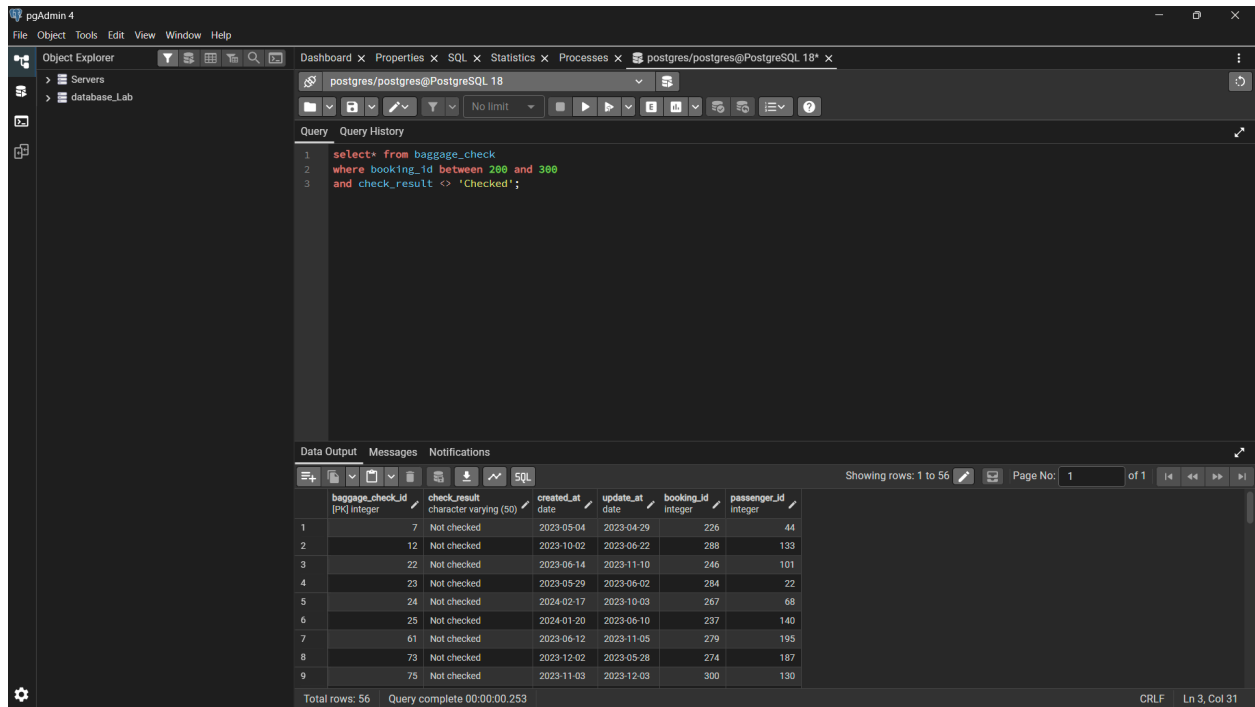
```

Data Output Messages Notifications

flight_no character varying (50)

Total rows: 0 Query complete 00:00:00.262 CRLF Ln 1, Col 17

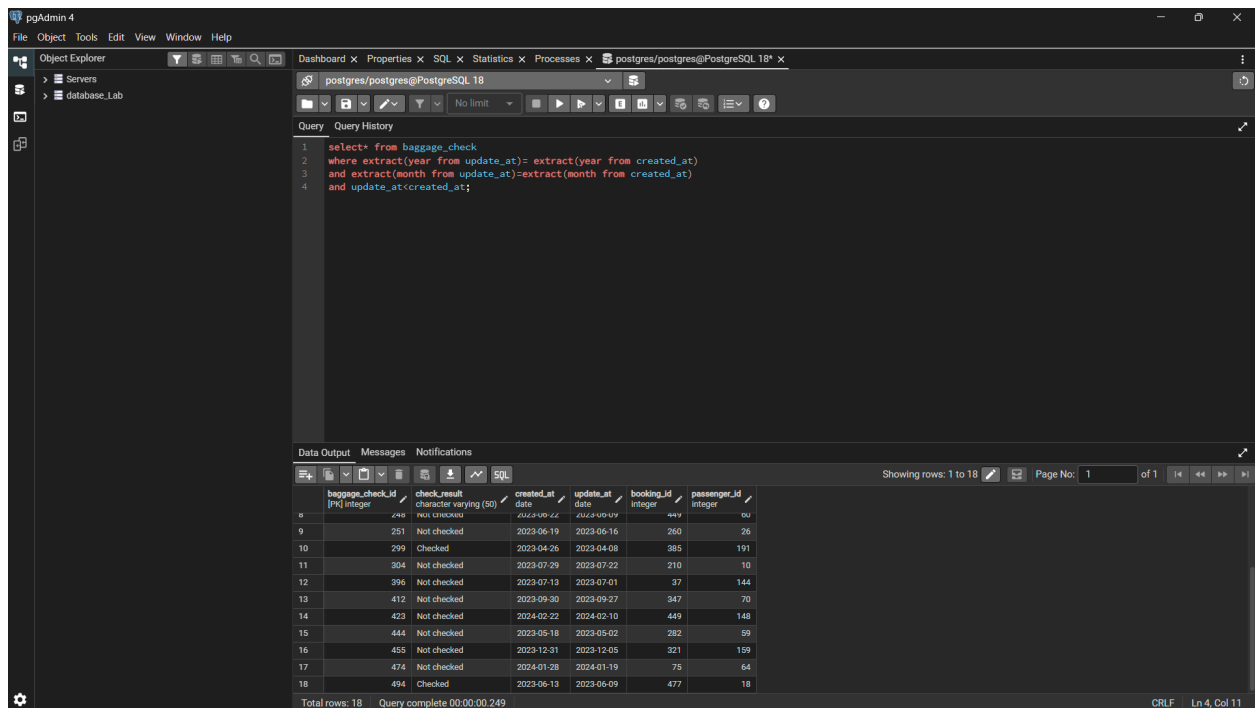
Task 14: Return all rows where booking_id is between 200 and 300 inclusive and check_result <> 'Checked'.



The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query filters for rows where booking_id is between 200 and 300 and check_result is not 'Checked'. The results table displays 9 rows of data.

baggage_check_id [PK] integer	check_result character varying (50)	created_at date	update_at date	booking_id integer	passenger_id integer
7	Not checked	2023-05-04	2023-04-29	226	44
12	Not checked	2023-10-02	2023-06-22	288	133
22	Not checked	2023-06-14	2023-11-10	246	101
23	Not checked	2023-05-29	2023-06-02	284	22
24	Not checked	2024-02-17	2023-10-03	267	68
25	Not checked	2024-01-20	2023-06-10	237	140
61	Not checked	2023-06-12	2023-11-05	279	195
73	Not checked	2023-12-02	2023-05-28	274	187
75	Not checked	2023-11-03	2023-12-03	300	130

Task 15: Baggage checks where update_at is in the same month as created_at but occurs earlier than created_at.



The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query filters for rows where the year and month of update_at match those of created_at, and update_at is earlier than created_at. The results table displays 18 rows of data.

baggage_check_id [PK] integer	check_result character varying (50)	created_at date	update_at date	booking_id integer	passenger_id integer
240	not checked	2023-09-22	2023-10-19	499	100
251	Not checked	2023-06-19	2023-06-16	260	26
299	Checked	2023-04-26	2023-04-08	385	191
304	Not checked	2023-07-29	2023-07-22	210	10
396	Not checked	2023-07-13	2023-07-01	37	144
412	Not checked	2023-09-30	2023-09-27	347	70
423	Not checked	2024-02-22	2024-02-10	449	148
444	Not checked	2023-05-18	2023-05-02	282	59
458	Not checked	2023-12-31	2023-12-05	321	159
474	Not checked	2024-01-28	2024-01-19	75	64
494	Checked	2023-06-13	2023-06-09	477	18

+ Homework

The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query is as follows:

```
1 select passenger_id, first_name, last_name, date_of_birth, extract(year from age(current_date, date_of_birth)) as age
2 from passenger
3 order by date_of_birth desc
4 limit 1;
```

The results are displayed in a table with the following columns: passenger_id, first_name, last_name, date_of_birth, and age. The table shows one row of data:

passenger_id	first_name	last_name	date_of_birth	age	
1	3	Algerim	Kalleva	1998-03-05	28

At the bottom of the interface, a green message box states: "Successfully run. Total query runtime: 194 msec. 1 rows affected." The status bar at the bottom indicates "Total rows: 1" and "Query complete 00:00:00.194".